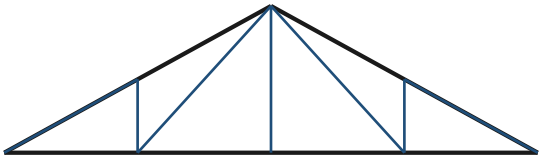
CE3201

DESIGN OF STEEL STRUCTURES



PROJECT REPORT

Analysis and Design of Roof Truss
for an Industrial Shed

Department of Civil Engineering

OBJECTIVE

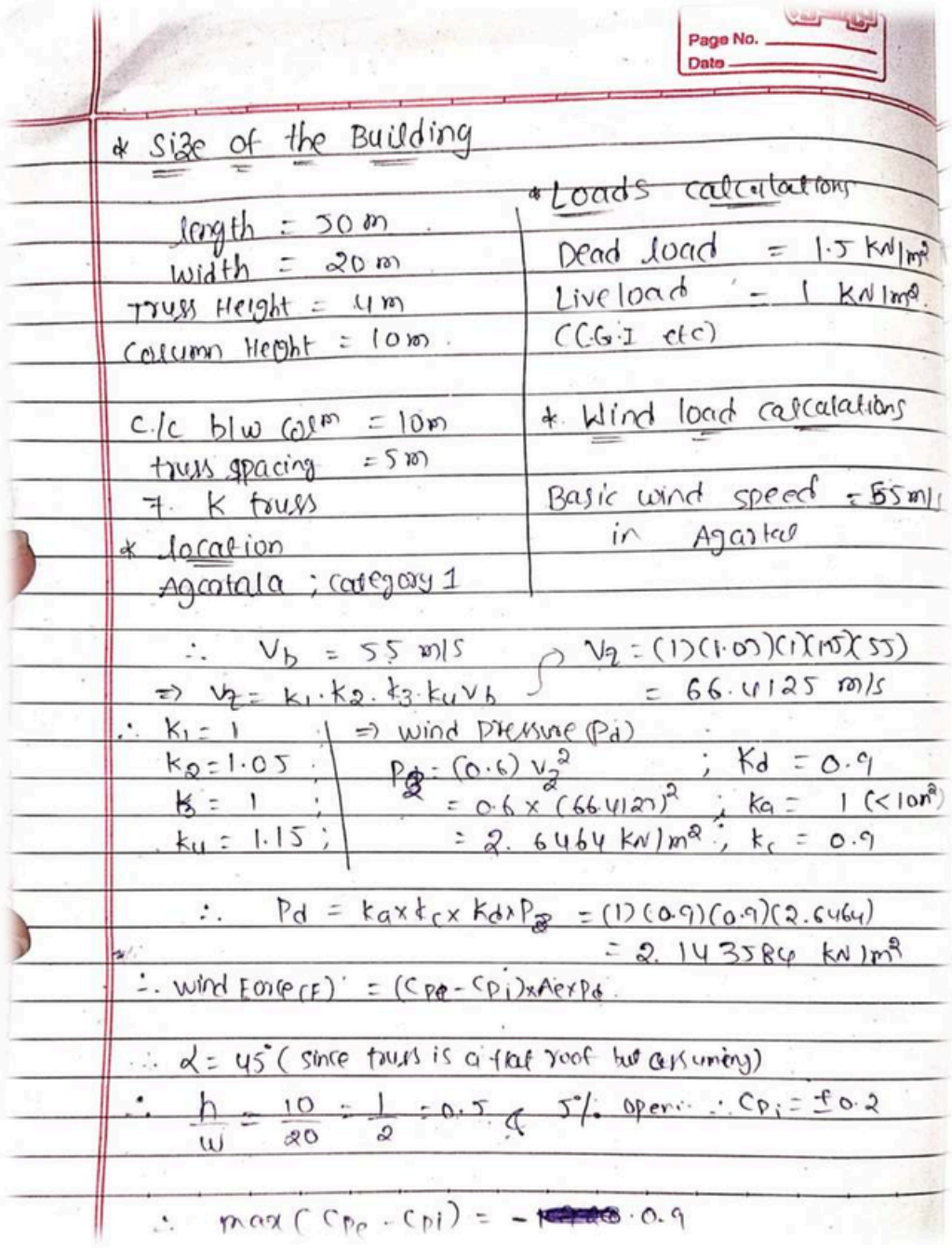
Designing an industrial shed roof truss with the following features:

- Proposed location of the industrial building to be designed for 50 years of life – Agartala, Tripura

- ***Dimensions of industrial shed –***

- Spacing of columns – 10 m (centre to centre)
- Ceiling Height – 10 m (shop floor to roof truss bottom)
- Yield strength of steel members = 250 MPa
- Spacing of trusses – not more than 5 m
- Roof span – 20m
- Roof angle – 45 degree
- Rise of the roof truss – approx. 4 m
- Truss type – K Truss Load cases:
- Dead load – 1.5 kpa
- Live load –1. kPa
- Wind load –To be calculated using IS875:Part III

WIND LOAD CALCULATION:



* Size of the Building

length = 30 m
width = 20 m
Truss Height = 4 m
Column Height = 10 m

c/c blw col^m = 10 m
truss spacing = 5 m
∴ K truss

* location
Agalawatta ; category 1

* Loads calculations

Dead load = 1.5 KN/m^2
Live load = 1 KN/m^2
(C.G.I etc)

* Wind load calculations

Basic wind speed = 55 m/s in Agalawatta

∴ V_b = 55 m/s

⇒ $V_z = k_1 \cdot k_2 \cdot k_3 \cdot k_4 V_b$

$V_z = (1)(1.05)(1)(1.15)(55)$
$= 66.4125$ m/s

∴ $k_1 = 1$
$k_2 = 1.05$
$k_3 = 1$
$k_4 = 1.15$

⇒ wind Pressure (P_d)

$P_d = (0.6) V_z^2$; $K_d = 0.9$
$= 0.6 \times (66.4125)^2$; $k_a = 1$ (<10m^2)
$= 2.6464$ KN/m^2 ; $k_c = 0.9$

∴ $P_d = k_a \times k_c \times k_d \times P_z = (1)(0.9)(0.9)(2.6464)$
$= 2.143584$ KN/m^2

∴ wind Force (F) = $(C_{pe} - C_{pi}) \times A_e \times P_d$

∴ $\alpha = 45°$ (since truss is a flat roof we assuming)

∴ $\frac{h}{w} = \frac{10}{20} = \frac{1}{2} = 0.5$ < 5% open ∴ $C_{pi} = \pm 0.2$

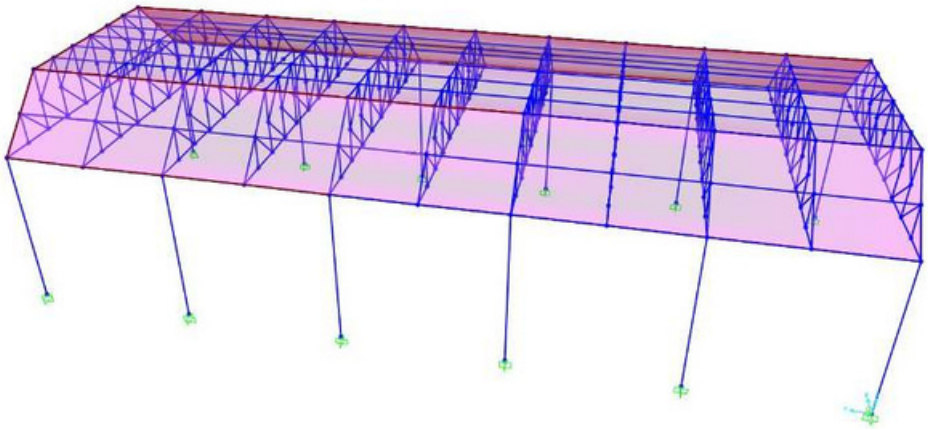
∴ max (C_{pe} - C_{pi}) = − [redacted] · 0.9

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Date

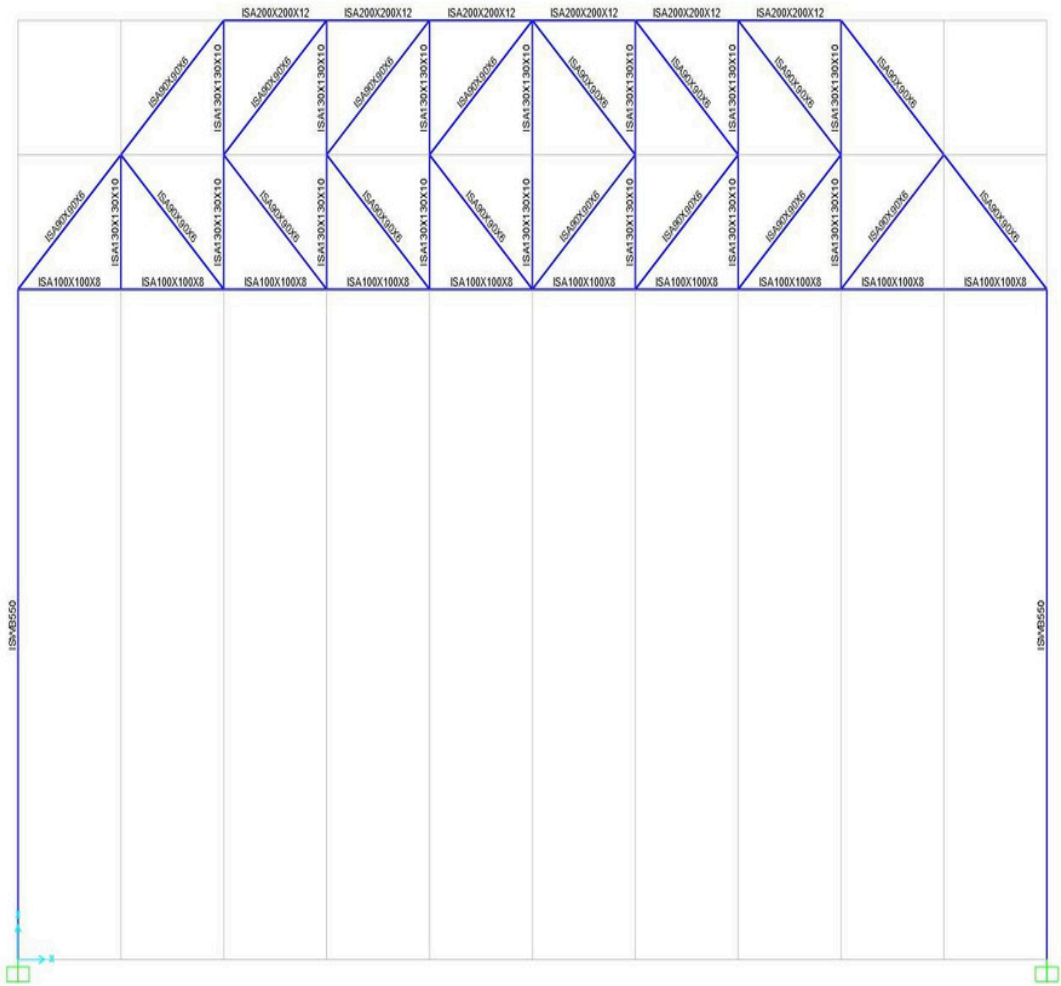
∴ $P = -0.9 \times P_d = -0.9 \times 2.143584$
$= -1.93$ KN/m^2

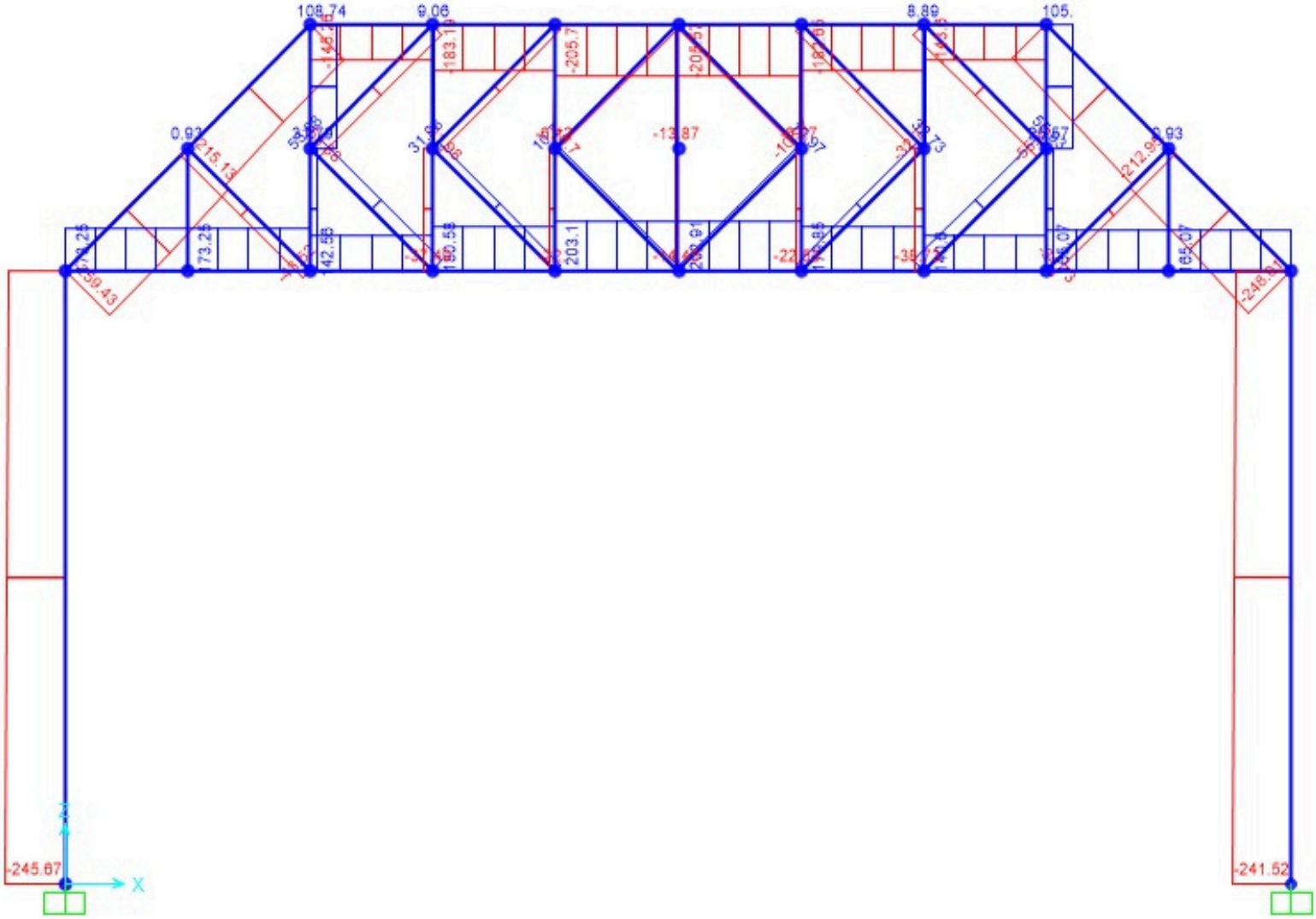
∴ wind load = 1.93 KN/m^2 (or) 1 kPa ∪.d.l.

3D VIEW OF STRUCTURE:



XZ VIEW OF PLANE:



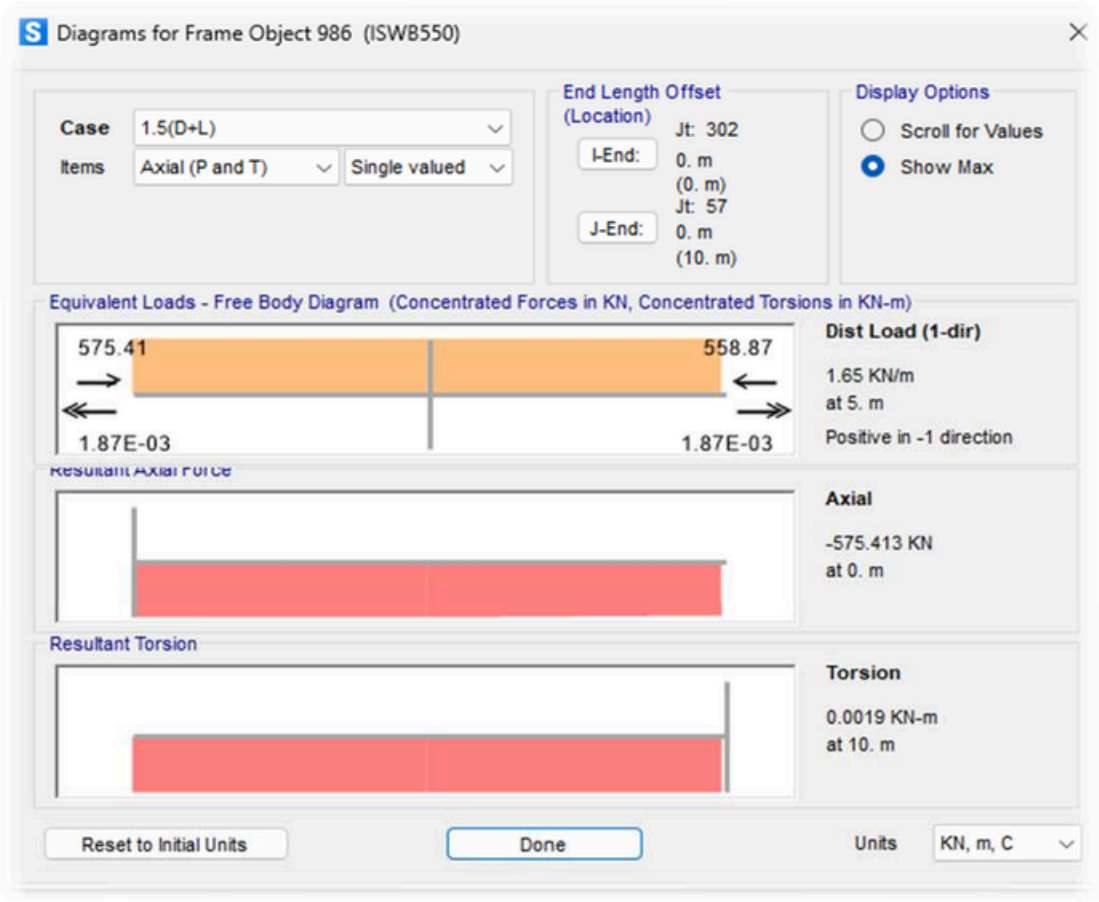


Axial force Diagram

ANALYSIS

COLUMN:

Max Axial Load: 575.61 KN ***(Load combination=1.5(D.L+L.L))***

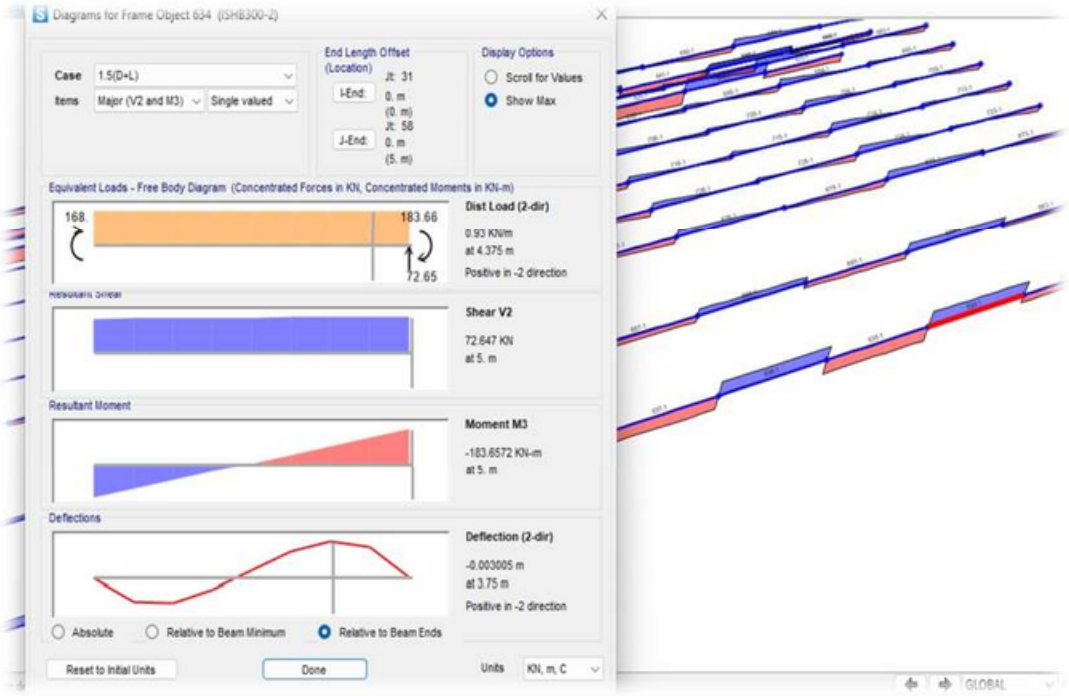


BEAM:

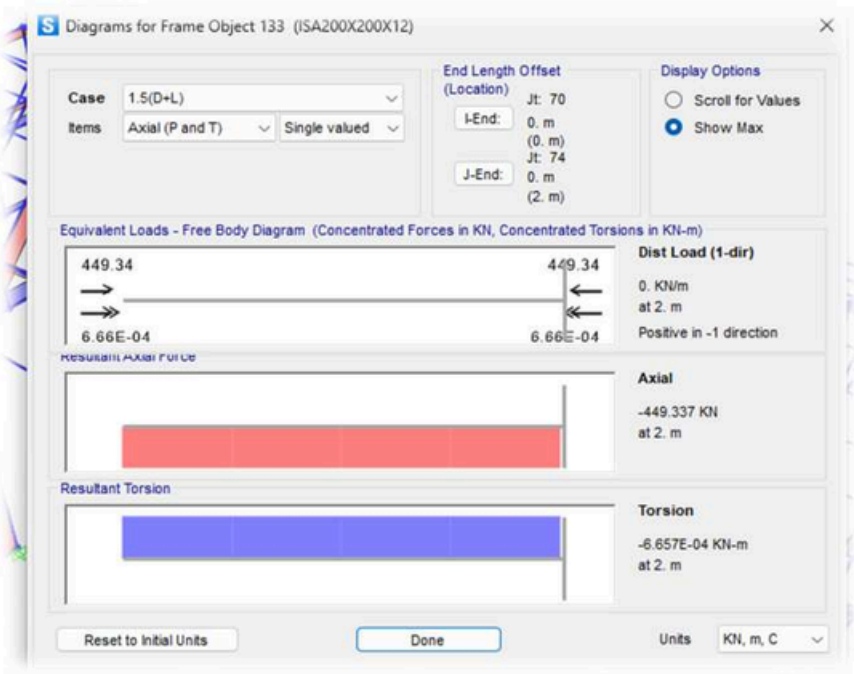
Max Moment = 183.657 KN

Max Shear Force = 72.647 KN

Load Combo = 1.5(D+L)

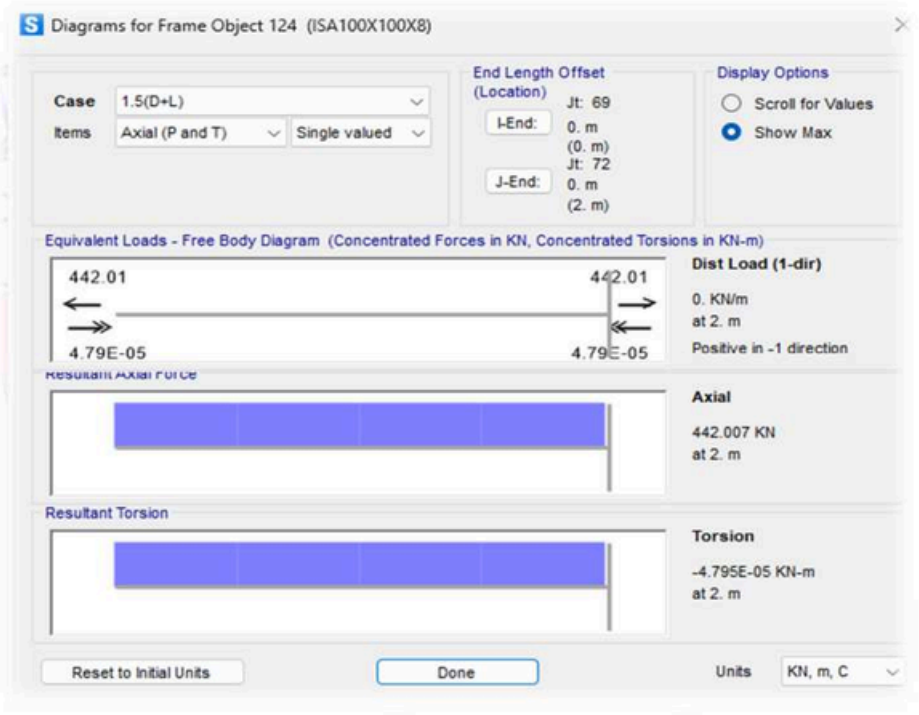


UPPER CHORD:



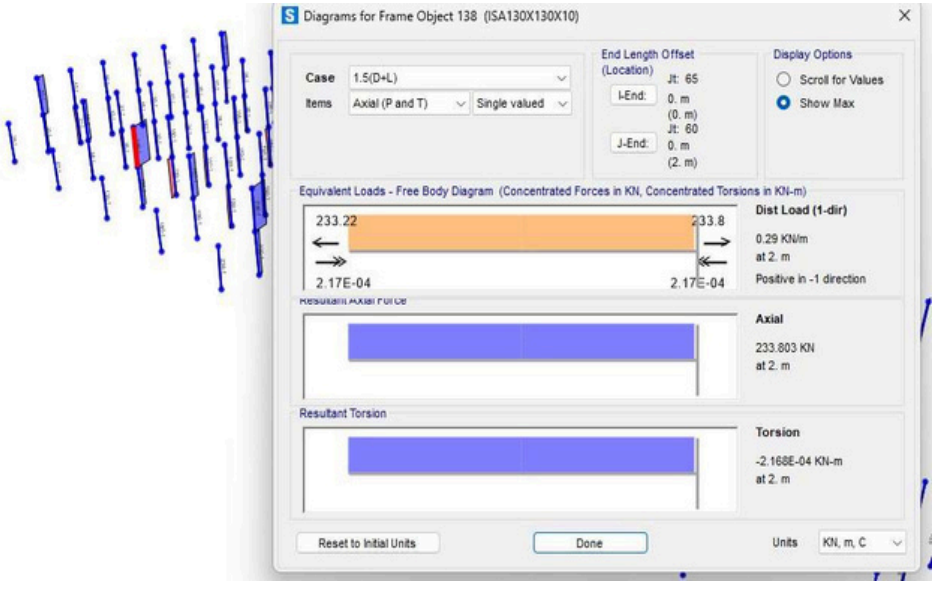
Max axial force = 449.34 (Compressive)

Bottom Chord :



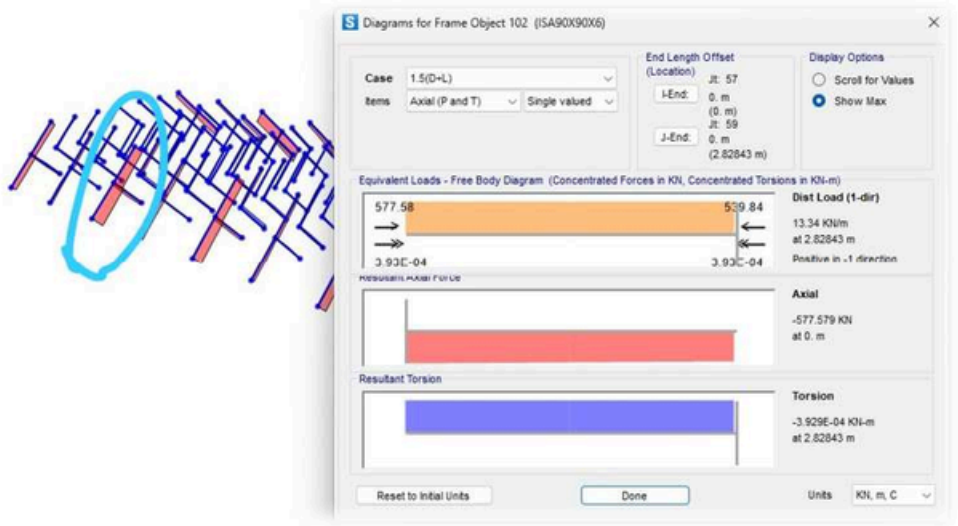
Max Axial Force = 446.49 (Tension)

Vertical Member:



Max Axial load: 233.22 (Tension)

Diagonal Member:



Max Axial Force =577.58 KN (compression)

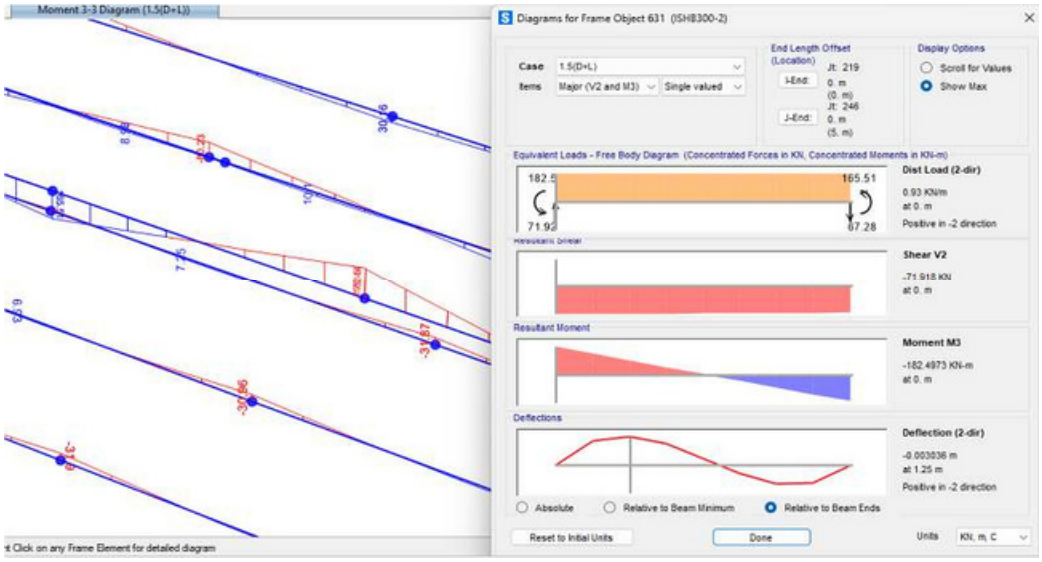
Purlins :

Max Moment in z direction M_z = 182.5 KNm

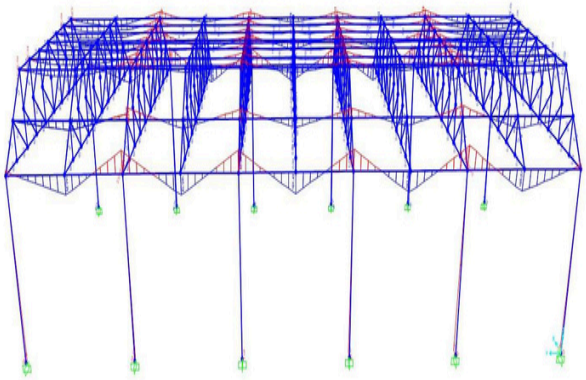
Max Moment in Y direction M_y = 7.99 KNm

Shear value in Z direction = 71.92 KN

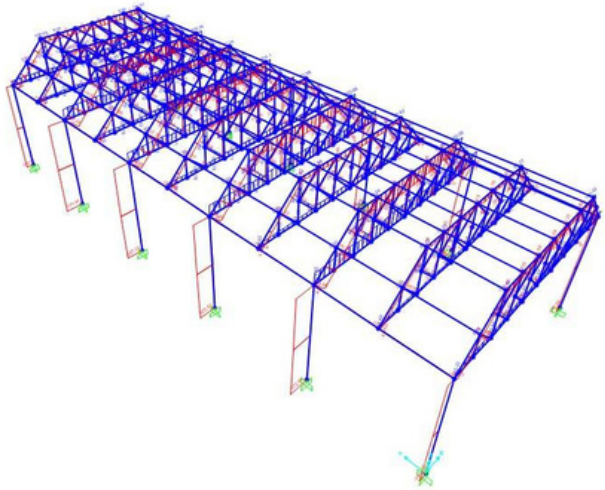
Shear value in Y direction = 2.633 KN



S.NO	*MEMBERS*	*LOAD COMB*	*AXIAL FORCE(KN)*	*MOMENT(KNm)*
1.	TOP CHORD	1.5(D+L)	449.34 (COMP)	-
2.	BOTTOM CHORD	1.5(D+L)	446.49 (TENSION)	-
3.	VERTICAL MEMBER	1.5(D+L)	233.22 (TENSION)	-
4.	DIAGONAL MEMBER	1.5(D+L)	577.58 (COMP)	-
5.	PURLIN	1.5(D+L)	- -	Mx= 182.5 My=7.99
6.	BEAM	1.5(D+L)	575.61 (COMP)	183.657
7.	COLUMN	1.5(D+L)		55.254



Bending Moment diagram



Axial Force Diagram

Design of Critical Column

Theaboveanalysis is done to thecritical column

We design the column for maximum load which we got from the analysis.
The column is designed as per Section 7 given in IS800.
We design column for compressive load

For Fe 410 grade of steel: f_y = 250Mpa

Partial safety factor: mo = 1.1, 1 = 1.25

Maximum compressive load =575.41
The length of the member =10 m
This load is the factored load i.e. 1.5(D.L + L.L) so there no need to multiply it with 1.5.
Hence, P_u = *575.41 KN*
As we already have the factored load lets calculate the area that would be required
For required area,
Let us assume slenderness ratio λ = 100 MPa
fcd = 118 MPa (Corresponding to Buckling Class b about y-y axis)

= ——— = .

Now, let us design the column member as a built-upsection.
Let us try ISMB 200 @ 249.1 N/m, 2 numbers. FromIS Handbook the relevant properties are

1.Sectionalarea,A=3232.67mm^2

2. Depth of section, h = 200mm

3. Width of Flange, b = 100mm

4. Thickness of flange, t_f= 10.8mm

5. Thickness of web, t_w = 5.7mm

6.MomentofInertia(z-zaxis),I=2235.4cm^4

7.MomentofInertia(y-yaxis),I=150.0cm^4

8. Root radius, r = 11mm

So,A_{prov} =2×3233=6466mm^2

1. Design compressive strength:

Slendernessratioforthis section = = ——— = ——————— =78.125)*+

Effectiveslendernessratio, ——=1.05∗78.125=82.03)*+; f_y=250Mpa

f_{cd}= 132.955 MPa, Therefore P_{cd} = 132.955×6466×10^{-3} = 860kn > 575.16 ***kN***

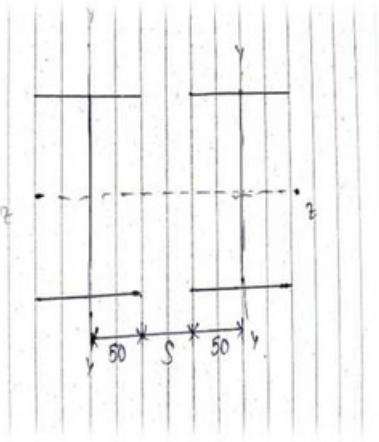
2. Spacing of I sections:

$2I_z = 2\{I_y + A(S/2 + C_{yy})^3\}$ □ $2\times2235.4\times10^4 = 2(150\times10 + 3233(S/2+50))^2$

S = 60 mm

3. Check for spacing:

From Clause 7.6.5.1



For lacing bars(45^{0}inclination)

a_1 = 2 (60+50+50)cot45^0=320

+ 320
1= 21.5=14.955067+8
2

0.7978.1=54.67 67+8

4. Shear capacity:

From Clause 7.6.6.1

Maximum Shear —— ∗ 57541 14.385;<

Compressive and tensile force in lacing= = > ?6@A? = B! ." ?6@A? 45°=10171.9 D

5. ***Dimension of lacing bar:***

From Clause 7.6.1

Diameter of bolt d_b= 20 mm

Width of lacing = $3d_b$ = 3 × 20 = 60 mm

Effective length of lacing bars L_e = (60+50+50) cosec 45^0 = 226 mm

Thickness of lacing bars = — B ∗ E. = "" B = 5.7 6 mm

Provide a flat section 226 ISF 6 mm

6. Design load for lacing:

From Clause 7.6.6.3

$1_{FG>}$ = $\overline{\sqrt{}}$ =1.732

= —— = $\frac{""}{.I!"}$ = 130.48 5 145 (okay)

For = 130.48 ; f_{cd}= 74.32 (Buckling class c)

P_{cd}= 74.32×60×6= 26.75 KN > 10.17 kn

7. Tensile Strength of plate:

a. $\frac{.J\ K\ LM^{\ N\ OPQ}}{RS\ \ T}$ = $\frac{.J\ \ L"\ *\ *B}{."}$ =70.848 ;D

b. $\frac{U\ vPW}{X}$ =60∗6∗ $\frac{"}{.}$ =81.8 ;D

Tensile Strength = min(a,b) = 70.848 KN

Hence, provide 2 ISMB 200 @ 249.2 N/m with spacing of 60mm and thickness of lacing bars 6mm

Designed by :-

K GANESHNAIK: 2301CE70

LIRE NIRIN : 2301CE57

Purlin Design

Factored Loads and Moments:

M_{ux} = 182.5 kNm

M_{uy} = 7.99 kNm

Taking Amplified Moment:

$M_{u'} = M_{ux} + k_b M_{uy}$

where (k_b = 4 to 10), taking k_b = 6

$M_{u'}$ = 182.5 + 6 × 7.99 = 230.44 kNm

$Z_{p,x}$ required = $(M_{u'} \times \gamma_{m0}) / f_y$ $= (230.44 \times 10^6 \times 1.1) / 250$

= 1,013,936 mm³ = 1,013.94 cm³

Trying Section: ISLB 400 @ 56.9 kg/m

Properties:

b_f = 165 mm	h = 400 mm
t_f = 12.5 mm	t_w = 8.0 mm
r = root radius = 16 mm	$Z_{p,x}$ = 1099.45 cm³
$Z_{e,x}$ = 965.3 cm³	$Z_{e,y}$ = 86.8 cm³

Section Classification:

b_f / t_f = 165/2 / 12.5 = 6.6 < 9.4ε

d / t_w = [400 - 2(12.5 + 16)] / 8 = 343 / 8 = 42.875 < 84ε

⇒ Plastic Section ⇒ β_b = 1.0

Design Bending Moment Capacity Check:

$M_{dx} = (1.0 \times 1099.45 \times 10^3 \times 250) / (1.1 \times 10^6)$ = 249.87 kNm

Limit check: $1.2 \times 965.3 \times 10^3 \times 250 / (1.1 \times 10^6)$ = 263.26 kNm

∴ 249.87 < 263.26 ⇒ M_{dx} = 249.87 kNm

$Z_{p,y} = (b_f^2 t_f / 2) + (d t_w^2 / 4)$ = 175.64 cm³

$M_{dy} = (1.0 \times 175.64 \times 10^3 \times 250) / (1.1 \times 10^6)$ = 39.92 kNm

Limit check: $1.2 \times 86.8 \times 10^3 \times 250 / (1.1 \times 10^6)$ = 26.04 kNm

∴ 39.92 > 26.04 ⇒ M_{dy} = 26.04 kNm (governed by limit)

Interaction Check:

$(M_{ux} / M dx) + (M_{uy} / M dy) = (182.5 / 249.87) + (7.99 / 26.04)$

=1.037 > 1 **(Not okay)**

We have to try for another section.

Trying Section: ISLB 450 @ 65.3 kg/m

h = 450 mm	tw = 8.6 mm
b f = 170 mm	r = root radius = 16 mm
t f = 13.4 mm	Z p,x = 1401.35 cm³
Z e,x = 1223.8 cm³	Z e,y = 100.4 cm³

$Z_{p,y} = (b_f t_f (2t/2) + ([h-2(r+t)]t_w^2 / 4)$

$= (1702 \times 13.4 / 2) + ([450 - 2(16 + 13.4)] \times 8.62 / 4)$

= 200.86 cm³

Section Classification:

b / t_f = 170/2 / 13.4 = 6.34 < 9.4

d / t_w = 391.2 / 8.6 = 45.49 < 84

⇒ Plastic Section ⇒ β_b = 1.0

Design Bending Capacities (M_{dx} and M_{dy}):

M_{dx} calculation: $(1.0 \times 1401.35 \times 10^3 \times 250) / (1.1 \times 10^6)$ = 318.49 kNm

Limit: $(1.2 \times 1223.8 \times 10^3 \times 250) / (1.1 \times 10^6)$ = 333.76 kNm

∴ M_{dx} = 318.49 kNm

$M_{dy,plastic}$ = $(1.0 \times 200.86 \times 10^3 \times 250) / (1.1 \times 10^6)$ = 45.65 kNm

Limit: 1.2 × 100.4 × 103 × 250 / (1.1 × 106) = 27.38 kNm

∴ M_{dy} = 27.38 kNm (governed by limit)

Interaction Check:

$(M_{ux} / M_{dx}) + (M_{uy} / M_{dy}) = (182.5 / 318.49) + (7.99 / 27.38)$

= 0.573 + 0.292 = 0.865 < 1 **(Safe)**

Shear Check:

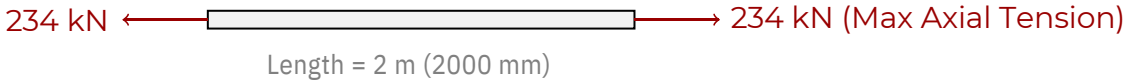
$V_d = (f_y h t_w) / (\sqrt{3} \gamma_{m0})$ = $(250 \times 450 \times 8.6) / (\sqrt{3} \times 1.1 \times 10^3)$ = 507.8 kN

$V_d > V_u$ **(Okay)**

Hence **ISLB 450 @ 65.3 kg/m** section is safe for the given loads.

Truss Member Design 1

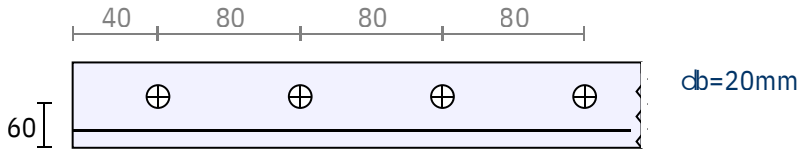
(Vertical Member - Tension member)



1. Design Parameters

Factored Load Calculation

Factored load (Axial) (T_u) = 234 × 1.5 = 351 kN



- $A_{n,req} = \frac{T_u}{0.8\left(\frac{f_u}{\gamma_{m1}}\right)} = \frac{351(\times 10^3}{0.8\ \frac{410}{})} = 1337.65\ mm^2$
- $A_{g,req} = \frac{T_u}{f_y/\gamma_{m0}} = \frac{351 \times 10^3}{250/1.1} = 1544.4\ mm^2$

2. Selection from Steel Table

Selected Section: ISA 100 × 65 × 10 @ 12.2 kg/m

Cross-sectional Area (Ag): 1551 mm^2

3. Strength Checks

Following IS 800: Section 6 - Design of Tension Members

A. Limit State of Yielding

$$T_{dg} = \frac{A_g \cdot f_y}{\gamma_{m0}} = \frac{1551 \times 250}{1.1} = 352.5kN > T_u \quad [OK]$$

B. Limit of Rupture

$$T_{dn} = \frac{0.9 \cdot A_{nc} \cdot f_u}{\gamma_{m1}} + \beta \frac{A_{go} \cdot f_y}{\gamma_{m0}}$$

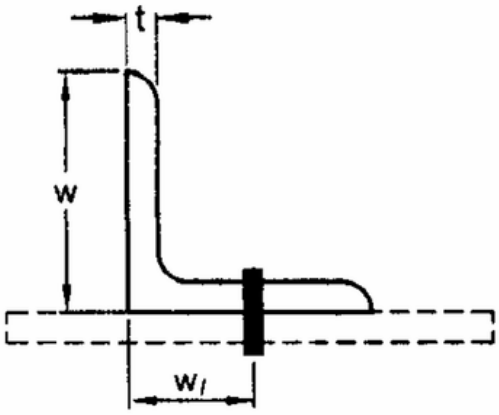
$b_s = 60 + 65 - 1(0 = 115mm$

$\beta = 1.4 - 0.076\ \frac{60}{}) (\ \frac{250}{}) (\ \frac{11}{5})$

$= 1.267 \quad \leq \frac{f_u \gamma_{m0}}{f_y \gamma_{m1}} = 1.44 \quad \frac{24}{0}$

≥ 0.7

So, b_s = 115 mm and β = 1.267.

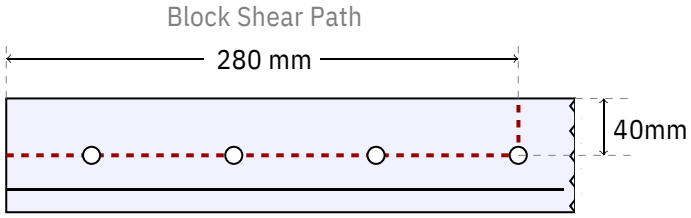


$$b_s = w + w_l - t$$

- $A_{nc} = (100 - 22 - 10/2) \times 10 = 730\ mm^2$
- $A_{go} = (65 - 10/2) \times 10 = 600 mm^2$

$$T_{dn} = \frac{0.9 \times 730 \times 410}{1.25} + \frac{1.267 \times 600 \times 250}{1.1} = 388.3\ kN > T_u \quad [OK]$$

4. Block Shear Analysis



$A_{tn} = 290\ mm^2$

$A_{vn} = 2030\ mm^2$

$A_{tg} = 400\ mm^2$

$A_{vg} = 2800\ mm^2$

$$T_{db1} = \frac{A_{tg} f_y}{\gamma_{m0}} + \frac{0.9 \times A_{vn} f_u}{\gamma_{m1} \sqrt{3}}$$

$$= \frac{400 \times 250}{1.1} + \frac{0.9 \times 203 \sqrt{0} \times 410}{1.25\ \sqrt{3}} = 436.9 kN$$

$$T_{db2} = \frac{0.9 A_{tn} f_u}{\gamma_{m1}} + \frac{A_{vg} \sqrt{f_y}}{\gamma_{m0}\ 3}$$

$$= \frac{0.9 \times 290 \times 410}{1.25} + \frac{2800 \times 250}{\sqrt{-}\ 1.13} = 453 kN$$

$T_{db} = \min(T_{db1}, T_{db2}) = 436.9\ kN > T_u$ [OK]

5. Final Summary & Serviceability

- Governing Strength (T_d): 352.5 kN

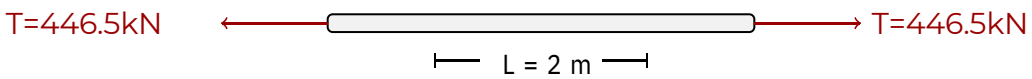
- Slenderness Ratio (λ): $\frac{2000}{16.5} = 121.21 < 350$ [OK]

Final Recommendation

Select ISA 100 × 65 × 10 @ 12.2 Kg/m for the vertical truss member. The section provides adequate safety against yielding, rupture, and block shear as per IS 800 standards.

Truss Member Design 2

(Bottom Chord - Tension member)

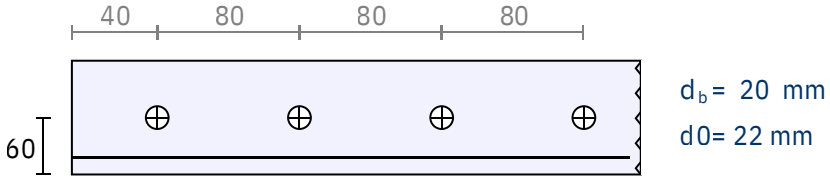


Maximum Axial Tensile load = 446.5 kN

1. Design Parameters

Factored Load Calculation

Factored load (T_u) = 446.5 × 1.5= 670 kN



- $A_{n,req} = \frac{T_u}{0.8\left(\frac{f_u}{\gamma_{m1}}\right)} = \frac{670(\times 10^3}{0.8\ \frac{410}{})} = 2526.7\ mm^2$

- $A_{g,req} = \frac{Tu}{fy\ /\gamma_{m0}} = \frac{670\times 10^3}{250/1.1} = 2917.2\ mm^2$

2. Selection from Steel Table

Selected Section: ISA 150 × 115 × 12 @ 23.8 kg/m
Cross-sectional Area (Ag): 3038 mm^2

3. Strength Checks

Following IS 800: Section 6 - Design of Tension Members

A. Limit State of Yielding

$$T_{dg} = \frac{A_g \cdot f_y}{\gamma_{m0}} = \frac{3038\times 250}{1.1} = 690.5kN > T_u \quad [OK]$$

B. Limit of Rupture

$$T_{dn} = \frac{0.9 \cdot A_{nc} \cdot f_u}{\gamma_{m1}} + \beta \frac{A_{go} \cdot f_y}{\gamma_{m0}}$$

$$b_s = 115 + 60 - (12 = 163\text{mm}$$

$$\beta = 1.4 - 0.076 \frac{115}{}) (\frac{250}{}) (\frac{163}{280})$$

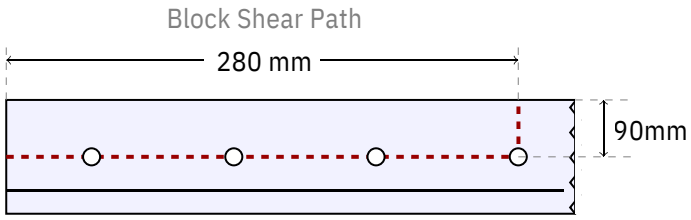
$$= 1.147 \leq \frac{f_u \gamma_{m0}}{f_y \gamma_{m1}} = 1.44$$

$$\geq 0.7$$

- $A_{nc} = (150 - 22 - 12/2) \times 12 = 1464\ mm^2$
- $A_{go} = (115 - 12/2) \times 12 = 1308\ mm^2$

$$T_{dn} = \frac{0.9 \times 1464 \times 410}{1.25} + \frac{1.147 \times 1308 \times 250}{1.1} = 773\ kN > T_u \quad [OK]$$

4. Block Shear Analysis



$A_{vg} = 280 \times 12 = 3360 mm^2$

$A_{tg} = 90 \times 12 = 1080 mm^2$

$A_{vn} = (280 - 3.5 \times 22) \times 12 = 2436\ mm^2$

$A_{tn} = (90 - 0.5 \times 22) \times 12 = 948\ mm^2$

$$T_{db1} = \frac{A_{tg} f_y}{\gamma_{m0}} + \frac{0.9 A_{vn} f_u}{\gamma_{m1} \sqrt{3}}$$

$$= \frac{1080 \times 250}{1.1} + \frac{0.9 \times 243 \sqrt{6} \times 410}{1.25 \sqrt{3}} = 672 kN$$

$$T_{db2} = \frac{0.9 A_{tn} f_u}{\gamma_{m1}} + \frac{A_{vg} \sqrt{f_y}}{\gamma_{m0} \sqrt{3}}$$

$$= \frac{0.9 \times 948 \times 410}{1.25} + \frac{3360 \times 250}{1.1 \times \sqrt{3}} = 720.7 kN$$

$$Tdb = \min(T_{db1}, T_{db2}) = 672\ kN > T_u \ [OK]$$

5. Final Summary & Limit State of Serviceability

- Governing Strength (T_d): $\min(T_{dg}, T_{dn}, T_{db}) =$ 672 kN $> T_u$ [OK]
- Slenderness Ratio (λ_{max}): $\frac{2000}{33.7} = 59.34 < 350$ [OK]

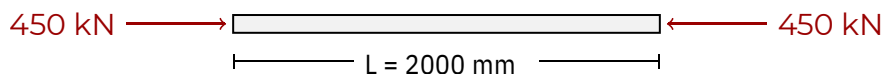
Final Recommendation

Select ISA 150 × 115 × 12 @ 23.8 kg/m for the bottom chord of the truss. The section provides adequate safety against yielding, rupture, and block shear, and satisfies serviceability limits.

Truss Member Design 3

(Upper Chord - Compression Member)

Fe 410 Steel



1. Design Parameters

Factored Compression Load Calculation

Factored Axial Load (P_u) = 450 × 1.5 = 675 kN

- Assumption: Let Slenderness Ratio (λ) ≈ 100.
- From IS 800 (Table 9c, Buckling Class 'c'), assumed f_{cd} = 107 MPa.
- $A_{g,req} = \frac{P_u}{f_{cd,assumed}} = \frac{675 \times 10^3}{107} = 6308\ mm^2$

2. Section Selection & Trial

Trial 1: Select ISA 200 × 200 × 18 ($A = 6881\ mm^2$). Checking this section:

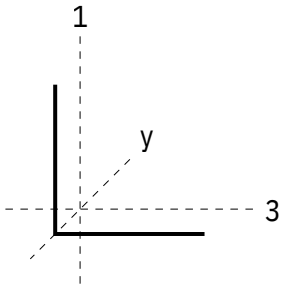
- $r_{min} = 61.3\ mm, \lambda = \frac{1 \times 2000}{61.3} = 32.63.$
- fcd = 208 MPa = $\Rightarrow P_{cd} = 208 \times 6881 = 1431\ kN \gg P_u$ (Oversized).

Trial 2 (Optimization): Try a lighter section for economy.

- Assumption: Let Slenderness Ratio (λ) ≈ 80.
- From IS 800 (Table 9c, Buckling Class 'c'), assumed f_{cd} = 136 MPa.

- $A_{g,req} = \frac{P_u}{f_{cd,assumed}} = \frac{675 \times 10^3}{136} = 4963\ mm^2$

Selected Section: ISA 150 × 150 × 18 @ 39.3 kg/m
Cross-sectional Area (Ag): 5079 mm^2



3. Compressive Strength Check

- Slenderness Ratio (λ):

$$\lambda = \frac{k \cdot L}{r_{min}} = \frac{1.0 \times 2000}{45.4} = 44.05$$

- Design Compressive Stress (f_{cd}): For λ = 44.05 and Buckling Class 'c' (IS800:Table 9c):

$$f_{cd} = 191\ MPa$$

- Design Compressive Strength (Pcd):

$$P_{cd} = f_{cd} \cdot A_g = 191 \times 5079 \times 10^{-3} = 970\ kN$$

P_{cd} = 970 kN > P_u = 675 kN [SAFE]

4. Final Recommendation

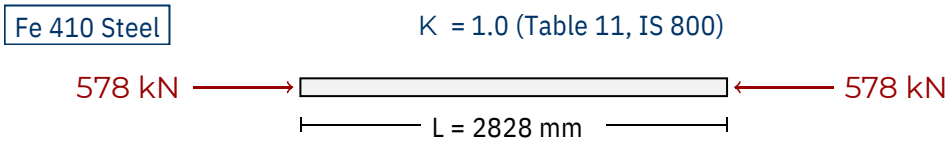
- Governing Strength: 970 kN
- Serviceability: λ = 44.05 < 180 (Limit for compression members).

Upper Chord Selection

Select ISA 150 × 150 × 18 @ 39.3 Kg/m for the upper chord. This section provides the lightest weight while maintaining a safety factor of 1.43 against compressive buckling.

Truss Member Design 4

(Diagonal Member - Compression)



1. Design Parameters

Factored Compression Load Calculation

Factored Axial Load (P_u) = 1.5 × 578 = 867 kN

- Assumption: Let assumed $\lambda = 80$.
- From IS 800 (Table 9c, Buckling Class 'c'), assumed $f_{cd} = 136$ MPa.
- Required Area:

$$A_{req} = \frac{P_u}{f_{cd,assumed}} = \frac{867 \times 10^3}{136} = 6375 \text{ mm}^2$$

2. Selection from Steel Table

Trial 1: Selected Section: ISA 200 × 200 × 18 @ 54 kg/m
Cross-sectional Area (A_g): 6881 mm^2
Minimum Radius of Gyration (rmin): 61.3 mm

3. Compressive Strength Check

- Actual Slenderness Ratio ( ):λ

$$\lambda = \frac{K.L}{r} = \frac{1 \times 2828}{61.3} = 46.13$$

- Design Compressive Stress (f_{cd}): Referring to IS 800, Table 9(c) for Buckling Class 'c':

Forλ = 46.13, f_{cd} = 188 MPa

- Design Compressive Strength (Pcd):

$$P_{cd} = f_{cd} \cdot A_g = \frac{188 \times 6881}{10^3} = 1294 \text{ kN}$$

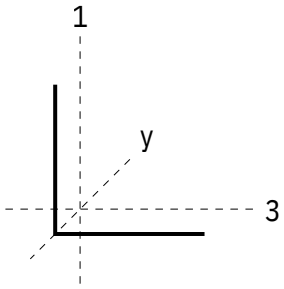
P_{cd} = 1294 kN > P_u = 867 kN [SAFE]

Trial 2 (Optimization): Try a lighter section for economy.

- Assumption: Let Slenderness Ratio (λ) ≈60.
- From IS 800 (Table 9c, Buckling Class 'c'), assumed f_{cd} = 168 MPa.

- $A_{g,req} = \frac{P_u}{f_{cd,assumed}} = \frac{867 \times 10^3}{168} = 5035\ mm^2$

Selected Section: ISA 150 × 150 × 18 @ 39.3 kg/m
Cross-sectional Area (Ag): 5079 mm^2



3. Compressive Strength Check

- Slenderness Ratio (λ):

$$\lambda = \frac{k \cdot L}{r_{min}} = \frac{1.0 \times 2000}{45.4} = 44.05$$

- Design Compressive Stress (f_{cd}): For λ = 44.05 and Buckling Class 'c' (IS800:Table 9c):

$$f_{cd} = 191\ MPa$$

- Design Compressive Strength (Pcd):

$$P_{cd} = f_{cd} \cdot A_g = 191 \times 5079 \times 10^{-3} = 970\ kN$$

P_{cd} = 970 kN > P_u = 867 kN [SAFE]

4. Final Summary

- Governing Strength: 970 kN
- Slenderness Limit Check: λ = 44.05 < 180 [OK]

Diagonal Member Selection

Select ISA 150 × 150 × 18 @ 39.3 Kg/m for the diagonal member. Analysis shows that this section is the lightest suitable option available that satisfies both compressive strength and slenderness requirements for the K-Truss.

Critical Beam Design

IS 800 : 2007(LimitStateMethod)|ISMB350(Trial)andISMB 550 (Adopted)

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1. Design Inputs

Max Bending Moment M	Max Shear Force V	Span L
183.657 kN·m	**72.647 kN**	**10 m**
Deflection Check Pointa	Permissible Deflection	Steel Grade
5 m (mid-span)	**L / 300 = 33.33 mm**	**Fe 250 (fy = 250 MPa)**

Material &Safety Parameters `f_y = 250MPa | γ0 = 1.10 | E = 2 x 10^5MPaEquivalent`

Design Equations Used

Moment Capacity:Md= $\frac{(fy*Zp)}{\gamma 0}$

Shear Capacity:Vd= $\frac{(fy*h*tw)}{\sqrt{3}*\gamma 0}$

Deflection (UDL): Δlim = $\frac{(P*a*(3l^2-4a^2)}{24*E*Ixx}$

Zp = $\frac{(Md*\gamma 0)}{fy}$

Zp=800.8*10 ^{3} mm^3

2. Trial Section — ISMB 350

Section Properties (IS 800 Annex H)

Property	Symbol	Value
Depth	H	350 mm
Flange width	Bf	140 mm
Flange thickness	Tf	14.2 mm
Web thickness	Tw	8.1 mm
Plastic section modulus	Zpxx	8,89,576.0 mm³
Elastic section modulus	Ze	889576 mm³
Moment of Inertia	Ixx	13,630.3 *10^4 mm^4 350
Web shear area	Aw = h x tw	x 8.1 = 2835 mm²
Root radius	R1	14mm

Step 1 section classification

Rolled Section :- $\frac{b}{tf} = \frac{140/2}{14.2} = 4.9295 < 9.8$

Rolled Section :- $\frac{d}{tw} = \frac{350-2(14.2)}{8.1} = 39.703 < 84$ $(So\ section\ is\ plastic)$

Step 2 — Bending Check

```
M_d = (f_y x Z_p) / gamma_m0
= (250 x 889576) / 1.10
```

= 202,175,000 N.mm = 202.175kN.m Applied M = 183.657 kN.m < M_d = 202.175 kN.m [PASS]

Step 3 — Shear Check

$Vd = \frac{[fy * h * tw)}{\sqrt{3} * \gamma 0}$

$= \frac{250*350*8.1}{\sqrt{3}*1.1}$

= 372 kN > 72.647 KN

Step 4 — Check web buckling

$\frac{d}{tw} = \frac{350-2(14.2)}{8.1}$ =39.703<67 (soneedforcheckforwen buckling)

Web buckling= $fw = (b1 + n1)tw * fck$ (b1=130 and n1=(h/2))

$\lambda = 2.5\frac{d1}{tw} = 2.5 * \frac{321.6}{8.1}$ =99.25 (fromclass c ,fcd=108.05)

fw=266.93KN>72KN

Step 5 — Check web cripping

Webcripping = $f = (b1 + n1)tw * fy$ (b1=130 and n1=2.5(tf+r1))

F=741.85kN > 72KN

Step 6 — Check web Delfection

$\Delta lim = \frac{(P*a*(3l^2 4a)^2}{24*E*Ixx}$

Δlim = 0.109m>0.033m

DEFLECTION CHECK FAILS — 109 mm > 33.33 mm (L/300). ISMB 350 is rejected. Revise to a deeper section.

Summary — ISMB 350

M_d (capacity)	V_d (capacity)	delta_actual
202.175 kN·m	**372 kN**	**109 mm**
Bending (M < M_d)		

	Calculated vs. Limit	Status
	183.657kN·m < 202.175 kN·m	PASS
Shear (V < 0.6 V_d)	72.629 kN < 372 kN	PASS
Deflection (delta < L/300)	109 mm > 33mm	FAIL

3. Adopted Section — ISMB 550

Section Properties (IS 808)

Property	Symbol	Value
Depth	H	550 mm
Flange width	Bf	190 mm
Flange thickness	Tf	19.3 mm
Web thickness	Tw	11.2 mm
Plastic section modulus	Zp	27,11,980 mm³
Elastic section modulus	Ze	2,359.8 cm³
Moment of Inertia	Ixx	64,893.6 cm
Web shear area	A_w = h x t_w	550 x 11.2 = 6160 mm²
Root radius	R1	18mm

Step 1 section classification

Rolled Section :- $\frac{b}{tf} = \frac{190/2}{19.3} = 4.9222 < 9.8$

Rolled Section :- $\frac{d}{tw} = \frac{550-2(19.3)}{11.2} = 45.66 < 84$ $(So\ section\ is\ plastic)$

Step 2 — Bending Check

M_d = (f_y x Z_p) / gamma_m0

= (250 x 27,11,980) / 1.10

= 6,16,35,90,90 N.mm = 616.35 kN.m Applied M = 182.657 kN.m < M_d = 616.35 kN.m

[PASS]

Step 3 — Shear Check

V_d = (f_y / sqrt(3)) x A_w / gamma_m0

= (250 / 1.732) x 6160 / 1.10

= 144.34 x 6160 / 1.10 = 8,09,290 N = 809.29 kN > 72.647 KN

Step 4 — Check web buckling

$\frac{d}{tw} = \frac{550-2(19.3)}{11.2} = 45.66 < 67$ (soneed for check for web buckling)

Web buckling= $fw = (b1 + n1)tw * fck$ (b1=130 and n1=(h/2))

$\lambda 1 = 2.5\frac{}{tw} = 2.5 * \frac{511.4}{11.2} = 114.15$ (from class c ,fcd=89.15)

fw=404.384KN>72.647KN

Step 5 — Check web cripping

Webcripping = $f = (b1 + n1)tw * fy$ (b1=130 and n1=2.5(tf+r1))

F=625.1kN > 72.647KN

Step 6 — Check lateral torsional buckling

Mcr ’=c1*Mcr C1=1.365 hf=550-19.3mm L=5000mm ry=37.3mm Iyy=1833.8cm^4

Mcr=$\frac{\pi 2 * E * Iy * hf}{2(L2)}(1+ \frac{1}{20}(\frac{(L/ry)}{(hf/tf)})^2)^{0.5}$ =568341703.2

Mcr’=775786424.9

Λlt=0.93

Φ=1.0091

Xlt=2.53

Fbd=575

Md=1559.938 KNm >182 KNm

Step 7 — Deflection Check

$\Delta lim = \frac{(P*a*(3l^2 - 4a)^2}{24*E*Ixx}$

Δlim = 0.22m<0.33m

ALL CHECKS SATISFIED — ISMB 550 is the adopted section.

Summary — ISMB 550

M_d (capacity)	V_d (capacity)	delta_actual
616.35 kN·m	**809.29 kN**	**22 mm**

	Calculated vs. Limit	Status
Bending (M < M_d)	182.657 kN·m < 616.33 kN·m	PASS

Shear (V < 0 V_d)	72.647 kN < 809.29 kN	PAS
Deflection (delta < L/300)	0.22 m < 0.33 m	S

			PAS

4. Design Comparison Summary

Check	ISMB 350	ISMB 550	Permissible
			S
Moment Capacity M_d	202.175 kN·m	616.33 kN·m	182.657 kN·m
Shear Capacity V_d	372 kN	809.29 kN	72.647 kN
Deflection delta_max	109 mm ✗	22 mm ✓	33.33 mm (L/300)
Overall Status	REJECTED	ADOPTED	—

5. Conclusion

Design Conclusion: Adopt ISMB 550

Final Recommendation: Provide ISMB 550 @ 1.074 kN/m over a 10 m simply supported span.

References: IS 800 : 2007 (General Construction in Steel — Code of Practice, LSM)

THANK YOU